Group I, Row III: explicit step-by-step simplification

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1 Result in one place

The four pieces N_1, N_2, N_{3a}, N_{3b} of Group I, Row III contain 22 terms. They reduce to

$$\frac{d^3 N_{\mathrm{I.III}}}{d^3 k} = \frac{1}{(2\pi)^3}\,\frac{i g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'} e^{-ik\cdot(w'-w)}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\mathcal{R}^{ij}\,\mathcal{W} \;+\; \text{c.c.}, \tag{1.1}$$

$$\mathcal{R}^{ij} = K^{ij}\left\{\log\frac{(x-z)^2}{(x-w)^2} + i\,\sigma\,\pi\right\} - \frac{1}{2}\,S^{ij}\,\log\frac{(z-w)^2}{(x-w)^2} + \mathcal{A}_0\,T^{ij}, \tag{1.2}$$

with the colour factor $\mathcal{W}$ of (2.1), the tensors K, S, T of (9.2) and $\sigma = \pm 1$ fixed by the $i\epsilon$ prescription. In two transverse dimensions this becomes (10.7), and the last two terms are a single one-parameter integral, (11.2).

2 Input and notation

2.1 What is common to all 22 terms

Every term of N_1, N_2, N_{3a}, N_{3b} contains

- the measure and phase $\dfrac{1}{(2\pi)^3}\displaystyle\int_{w,w'} e^{-ik\cdot(w'-w)} \int_{x,y,z,x'}$ and the factor $\dfrac{1}{k^+}$;
- the prefactor $\dfrac{i g^4 f^{abd}}{8\pi^5}$, or $\dfrac{i g^4 f^{abd}}{16\pi^5} = \dfrac{1}{2}\cdot\dfrac{i g^4 f^{abd}}{8\pi^5}$;

- the two vectors $\dfrac{(x'-w')}{(x'-w')^2}$ and $\dfrac{(y-z)}{(y-z)^2}$, each contracted with something;
- the same Wilson-line factor

$$\mathcal{W} \equiv \Big[U^{ab'}(x') - U^{ab'}(w')\Big]\rho^{b'}(x')\; f^{abd}\, U^{de}(x)\Big[U^{bc}(z) - U^{bc}(y)\Big]\rho^{e}(x)\,\rho^{c}(y). \tag{2.1}$$

$\mathcal{W}$ does not depend on w.

2.2 Notation

$$\begin{aligned} &a \equiv x-w, \qquad b \equiv x-z, \qquad c \equiv z-w = a-b, \qquad A \equiv a^2, \quad B \equiv b^2, \quad C \equiv c^2 = A+B-2\,a\cdot b, \\ &u^i \equiv \frac{(x'-w')^i}{(x'-w')^2}, \qquad v^j \equiv \frac{(y-z)^j}{(y-z)^2}. \end{aligned} \tag{2.2}$$

Identities used below:

$$c\cdot a = A - a\cdot b, \qquad c\cdot b = a\cdot b - B, \qquad A+B-C = 2\,a\cdot b. \tag{2.3}$$

The four logarithms and the function $\mathcal{A}_0$ that occur are

$$L_a \equiv \log\frac{B}{B-A} = \log\frac{-B}{A-B}, \qquad L_b \equiv \log\frac{A-B}{A}, \qquad L_c \equiv \log\frac{-B}{A}, \qquad L_d \equiv \log\frac{C}{A}, \tag{2.4}$$

$$\mathcal{A}_0 = \frac{1}{\Delta}\left\{\arctan\frac{B - a\cdot b}{\Delta} + \arctan\frac{a\cdot b}{\Delta}\right\}, \qquad \Delta \equiv \sqrt{AB-(a\cdot b)^2}. \tag{2.5}$$

L_a is the logarithm of the first term of N_1 and of all six terms of N_2 (written there as $\log[-(x-z)^2/((x-w)^2-(x-z)^2)]$). L_b is the logarithm of the second term of N_1 and of the first six terms of N_{3a}. L_c occurs once, in the third term of N_1. L_d occurs in the last three terms of N_{3b}.

2.3 Definition of the kernel

Writing each term as $u^i v^j \times (\text{tensor})^{ij}$, Row III is

$$\frac{d^3N_{\text{I.III}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'} e^{-ik\cdot(w'-w)}\; u^i\, v^j\; \mathcal{R}^{ij}\; \mathcal{W} + \text{c.c.}, \qquad \mathcal{R}^{ij} = \sum_{n=1}^{22} t_n^{ij}. \tag{2.6}$$

Example of the translation: the first term of N_2 is

$$-\frac{ig^4 f^{abd}}{8\pi^5}\,\log\left[\frac{-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\;(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \;\longrightarrow\; t^{ij} = -L_a\,\frac{c^i a^j}{C\,(A-B)}, \tag{2.7}$$

where the overall minus sign is the sign in front of the braces of N_2.

3 Step 1: the 22 terms

In units of the prefactor of (2.6):

N_1 (three terms).

$$t_1 = +\frac{1}{2}\,\delta^{ij}\,\frac{A+B}{(B-A)^2}\,L_a, \qquad t_2 = +\frac{1}{2}\,\delta^{ij}\,\frac{A+B}{(B-A)^2}\,L_b, \qquad t_3 = -\frac{a^i b^j}{AB}\,L_c. \tag{3.1}$$

N_2 (six terms, overall sign $-$, all with L_a).

$$\begin{aligned} &t_4 = -L_a\,\frac{c^i a^j}{C\,(A-B)}, \qquad t_5 = +L_a\,\frac{B}{A}\,\frac{a^i c^j}{C\,(A-B)}, \qquad t_6 = -L_a\,\delta^{ij}\,\frac{(c\cdot a)\,B}{C\,(A-B)^2}, \\ &t_7 = +L_a\,\frac{A}{B}\,\frac{c^i b^j}{C\,(A-B)}, \qquad t_8 = -L_a\,\frac{b^i c^j}{C\,(A-B)}, \qquad t_9 = +L_a\,\delta^{ij}\,\frac{(c\cdot b)\,A}{C\,(A-B)^2}. \end{aligned} \tag{3.2}$$

N_{3a} **(nine terms).**

$$t_{10} = -L_b\,\delta^{ij}\,\frac{(c\cdot a)\,B}{C\,(A-B)^2}, \qquad t_{11} = +L_b\,\delta^{ij}\,\frac{(c\cdot b)\,A}{C\,(A-B)^2}, \qquad t_{12} = -L_b\,\frac{c^i a^j}{C\,(A-B)},$$
$$t_{13} = +L_b\,\frac{A}{B}\,\frac{c^i b^j}{C\,(A-B)}, \qquad t_{14} = +L_b\,\frac{B}{A}\,\frac{a^i c^j}{C\,(A-B)}, \qquad t_{15} = -L_b\,\frac{b^i c^j}{C\,(A-B)},$$
$$t_{16} = +\mathcal{A}_0\,\frac{c^i a^j}{C}, \qquad t_{17} = +\mathcal{A}_0\,\frac{b^i c^j}{C}, \qquad t_{18} = -\mathcal{A}_0\,\frac{(a\cdot b)\,c^i b^j}{C\,B}. \tag{3.3}$$

N_{3b} **(four terms).**

$$t_{19} = -\mathcal{A}_0\,\frac{(a\cdot b)\,a^i c^j}{C\,A}, \qquad t_{20} = -\frac{1}{2}\,L_d\,\frac{c^i b^j}{C\,B}, \qquad t_{21} = +\frac{1}{2}\,L_d\,\frac{a^i c^j}{C\,A}, \qquad t_{22} = -\frac{1}{2}\,L_d\,\frac{\delta^{ij}}{C}. \tag{3.4}$$

4 Step 2: group by logarithm

function	terms
L_a	$t_1,\ t_4, \ldots, t_9$
L_b	$t_2,\ t_{10}, \ldots, t_{15}$
L_c	t_3
L_d	t_{20}, t_{21}, t_{22}
$\mathcal{A}_0$	$t_{16}, t_{17}, t_{18}, t_{19}$

Comparing $t_4, \ldots, t_9$ with $t_{12}, t_{14}, t_{10}, t_{13}, t_{15}, t_{11}$ one sees that they are pairwise identical up to $L_a \leftrightarrow L_b$. Define

$$M^{ij} \equiv \frac{1}{C}\left\{\frac{c^i a^j + b^i c^j}{A-B} - \frac{1}{A-B}\left[\frac{B}{A}\,a^i c^j + \frac{A}{B}\,c^i b^j\right] + \delta^{ij}\,\frac{B\,(c\cdot a) - A\,(c\cdot b)}{(A-B)^2}\right\}. \tag{4.1}$$

Then $t_4 + \cdots + t_9 = -M^{ij} L_a$ and $t_{10} + \cdots + t_{15} = -M^{ij} L_b$, and the coefficients of L_a and L_b are *the same tensor*:

$$\mathcal{R}^{ij} = \mathcal{Q}^{ij}\,\big(L_a + L_b\big) - \frac{a^i b^j}{AB}\,L_c + \big(t_{20} + t_{21} + t_{22}\big) + \big(t_{16} + \cdots + t_{19}\big), \qquad \mathcal{Q}^{ij} \equiv \frac{\delta^{ij}}{2}\,\frac{A+B}{(A-B)^2} - M^{ij}. \tag{4.2}$$

5 Step 3: the tensor $\mathcal{Q}^{ij}$ — the poles at $A = B$ cancel

δ^{ij} **part.** With (2.3),

$$B\,(c\cdot a) - A\,(c\cdot b) = B\,(A - a\cdot b) - A\,(a\cdot b - B) = 2AB - (A+B)\,a\cdot b. \tag{5.1}$$

Hence the coefficient of δ^{ij} in $\mathcal{Q}^{ij}$ is

$$\begin{aligned}\frac{A+B}{2(A-B)^2} - \frac{2AB - (A+B)\,a\cdot b}{C\,(A-B)^2} &= \frac{(A+B)\,C - 4AB + 2(A+B)\,a\cdot b}{2\,C\,(A-B)^2} \\ &= \frac{(A+B)(A+B-2a\cdot b) - 4AB + 2(A+B)\,a\cdot b}{2\,C\,(A-B)^2} = \frac{(A+B)^2 - 4AB}{2\,C\,(A-B)^2} = \frac{1}{2C}.\end{aligned} \tag{5.2}$$

The double pole $1/(A-B)^2$ has cancelled.

Tensor part. Substitute $c = a - b$:

$$\begin{aligned} c^i a^j + b^i c^j &= (a^i a^j - b^i a^j) + (b^i a^j - b^i b^j) = a^i a^j - b^i b^j, \\ \frac{B}{A}\,a^i c^j + \frac{A}{B}\,c^i b^j &= \frac{B}{A}\,(a^i a^j - a^i b^j) + \frac{A}{B}\,(a^i b^j - b^i b^j). \end{aligned} \tag{5.3}$$

The difference is

$$a^i a^j\left(1 - \frac{B}{A}\right) - b^i b^j\left(1 - \frac{A}{B}\right) + a^i b^j\left(\frac{B}{A} - \frac{A}{B}\right) = (A-B)\left[\frac{a^i a^j}{A} + \frac{b^i b^j}{B} - \frac{(A+B)\,a^i b^j}{AB}\right], \tag{5.4}$$

so the single pole $1/(A-B)$ cancels as well:

$$M^{ij}\Big|_{\text{tensor}} = \frac{1}{C}\left[\frac{a^i a^j}{A} + \frac{b^i b^j}{B} - \frac{(A+B)\,a^i b^j}{AB}\right]. \tag{5.5}$$

Result.

$$\mathcal{Q}^{ij} = \frac{1}{C}\left\{\frac{\delta^{ij}}{2} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{(A+B)\,a^i b^j}{AB}\right\}. \tag{5.6}$$

$\mathcal{Q}^{ij}$ is regular at $(x-w)^2 = (x-z)^2$.

6 Step 4: the logarithms with sign-indefinite arguments

	region $A > B$	region $A < B$
$L_a = \log\dfrac{B}{B-A}$ $L_b = \log\dfrac{A-B}{A}$	argument < 0: $\log\dfrac{B}{A-B} + i\sigma\pi$ argument > 0: $\log\dfrac{A-B}{A}$	argument > 0: $\log\dfrac{B}{B-A}$ argument < 0: $\log\dfrac{B-A}{A} + i\sigma\pi$
$L_a + L_b$	$\log\dfrac{B}{A} + i\sigma\pi$	$\log\dfrac{B}{A} + i\sigma\pi$

In each region exactly one of the two logarithms has a negative argument, $\log|A-B|$ cancels in the sum, and

$$L_a + L_b = L_c = \log\frac{B}{A} + i\,\sigma\,\pi \qquad \text{for all } A, B. \tag{6.1}$$

Here $\log(-|t|) = \log|t| + i\sigma\pi$ with one common $\sigma = \pm 1$ for all three logarithms ($\sigma = +1$ is the principal branch); this is an assumption about the $i\epsilon$ prescription of the original integrals.

Inserting (6.1) and (5.6) in (4.2), the three logarithms L_a, L_b, L_c multiply

$$\mathcal{Q}^{ij} - \frac{a^i b^j}{AB} = \frac{1}{C}\left\{\frac{\delta^{ij}}{2} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{(A+B-C)\,a^i b^j}{AB}\right\} = \frac{1}{C}\left\{\frac{\delta^{ij}}{2} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{2\,(a\cdot b)\,a^i b^j}{AB}\right\} \equiv K^{ij}, \tag{6.2}$$

so that

$$t_1 + \cdots + t_{15} = K^{ij}\left\{\log\frac{B}{A} + i\sigma\pi\right\}. \tag{6.3}$$

7 Step 5: the L_d terms

With $c = a - b$,

$$t_{20} + t_{21} + t_{22} = -\frac{L_d}{2C}\left[\frac{c^i b^j}{B} - \frac{a^i c^j}{A} + \delta^{ij}\right] = -\frac{L_d}{2C}\left[\frac{a^i b^j - b^i b^j}{B} - \frac{a^i a^j - a^i b^j}{A} + \delta^{ij}\right] = -\frac{1}{2}\,L_d\,S^{ij}, \tag{7.1}$$

$$S^{ij} \equiv \frac{1}{C}\left\{\delta^{ij} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{(A+B)\,a^i b^j}{AB}\right\} = \mathcal{Q}^{ij} + \frac{\delta^{ij}}{2C} = K^{ij} + \frac{\delta^{ij}}{2C} + \frac{a^i b^j}{AB}. \tag{7.2}$$

8 Step 6: the $\mathcal{A}_0$ terms

8.1 The tensor

$$\begin{aligned} t_{16} + \cdots + t_{19} &= \frac{\mathcal{A}_0}{C}\left\{c^i a^j + b^i c^j - (a\cdot b)\left[\frac{c^i b^j}{B} + \frac{a^i c^j}{A}\right]\right\} \\ &= \frac{\mathcal{A}_0}{C}\left\{a^i a^j - b^i b^j - (a\cdot b)\left[\frac{a^i b^j - b^i b^j}{B} + \frac{a^i a^j - a^i b^j}{A}\right]\right\} \\ &= \frac{\mathcal{A}_0}{C}\left\{\frac{A - a\cdot b}{A}\,a^i a^j + \frac{a\cdot b - B}{B}\,b^i b^j - (a\cdot b)\,\frac{A-B}{AB}\,a^i b^j\right\} = \mathcal{A}_0\,T^{ij}, \end{aligned} \tag{8.1}$$

$$T^{ij} \equiv \frac{1}{C}\left\{\frac{(c\cdot a)}{A}\,a^i a^j + \frac{(c\cdot b)}{B}\,b^i b^j - \frac{(a\cdot b)\,(A-B)}{AB}\,a^i b^j\right\}. \tag{8.2}$$

8.2 The function $\mathcal{A}_0$

$\Delta = \sqrt{AB - (a\cdot b)^2} = |a|\,|b|\sin\theta_x$ is twice the area of the triangle with corners x, z, w; θ_x is its interior angle at x (between a and b), θ_z the one at z (between $x - z = b$ and $w - z = -c$), θ_w the one at w.

- $\dfrac{a\cdot b}{\Delta} = \cot\theta_x$, so $\arctan\dfrac{a\cdot b}{\Delta} = \dfrac{\pi}{2} - \theta_x$.
- $B - a\cdot b = b\cdot(-c) = |b|\,|c|\cos\theta_z$ and $\Delta = |b\times c| = |b|\,|c|\sin\theta_z$ (because $a\times b = (b+c)\times b = c\times b$), so $\arctan\dfrac{B - a\cdot b}{\Delta} = \dfrac{\pi}{2} - \theta_z$.

Therefore

$$\mathcal{A}_0 = \frac{\pi - \theta_x - \theta_z}{\Delta} = \frac{\theta_w}{\Delta}, \qquad \cos\theta_w = \frac{(x-w)\cdot(z-w)}{|x-w|\,|z-w|}, \quad 0 < \theta_w < \pi. \tag{8.3}$$

Equivalently, since $\int_0^1 \frac{dt}{Bt^2 - 2(a\cdot b)t + A} = \frac{1}{\Delta}\Big[\arctan\frac{Bt - a\cdot b}{\Delta}\Big]_0^1$,

$$\mathcal{A}_0 = \int_0^1 \frac{dt}{(a - t\,b)^2} = \int_0^1 \frac{dt}{\big(x_t - w\big)^2}, \qquad x_t \equiv (1-t)\,x + t\,z, \tag{8.4}$$

the integral of the inverse squared distance from w to the points of the segment between x and z. $\mathcal{A}_0$ is real and has no logarithm.

9 Step 7: result

Adding Steps 4–6,

$$\boxed{\mathcal{R}^{ij} = K^{ij}\left\{\log\frac{(x-z)^2}{(x-w)^2} + i\,\sigma\,\pi\right\} - \frac{1}{2}\,S^{ij}\,\log\frac{(z-w)^2}{(x-w)^2} + \mathcal{A}_0\,T^{ij}} \tag{9.1}$$

with, in the original variables,

$$\begin{aligned}
K^{ij} &= \frac{1}{(z-w)^2}\left\{\frac{\delta^{ij}}{2} - \frac{(x-w)^i(x-w)^j}{(x-w)^2} - \frac{(x-z)^i(x-z)^j}{(x-z)^2} + \frac{2\,(x-w)\cdot(x-z)\;(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2}\right\},\\
S^{ij} &= \frac{1}{(z-w)^2}\left\{\delta^{ij} - \frac{(x-w)^i(x-w)^j}{(x-w)^2} - \frac{(x-z)^i(x-z)^j}{(x-z)^2} + \frac{\big[(x-w)^2 + (x-z)^2\big]\,(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2}\right\},\\
T^{ij} &= \frac{1}{(z-w)^2}\left\{\frac{(z-w)\cdot(x-w)}{(x-w)^2}\,(x-w)^i(x-w)^j + \frac{(z-w)\cdot(x-z)}{(x-z)^2}\,(x-z)^i(x-z)^j\right.\\
&\qquad\left. - \frac{(x-w)\cdot(x-z)\,\big[(x-w)^2 - (x-z)^2\big]}{(x-w)^2(x-z)^2}\,(x-w)^i(x-z)^j\right\}.
\end{aligned} \tag{9.2}$$

What happened to the 22 terms:

- $t_1, \ldots, t_{15}$ (three different logarithms, single and double poles at $A = B$) $\to$ one term $K^{ij}[\log(B/A) + i\sigma\pi]$, no pole;
- $t_{20}, t_{21}, t_{22} \to -\frac{1}{2}S^{ij}\log(C/A)$;
- $t_{16}, \ldots, t_{19} \to \mathcal{A}_0 T^{ij}$.

K^{ij} is the bracket of Group I Row IV, Group II Row I ($1a$) and Group III Rows I and II: with $c = a - b$, $\frac{b^j c^i}{CB} - \frac{a^i c^j}{CA} - \frac{a^i b^j}{AB} + \frac{\delta^{ij}}{2C} = K^{ij}$.

10 Step 8: further simplification in two transverse dimensions

Let θ be the signed angle from $a = x - w$ to $b = x - z$,

$$\cos\theta = \frac{a\cdot b}{|a|\,|b|}, \qquad \sin\theta = \frac{a\times b}{|a|\,|b|}, \qquad a\times b \equiv \epsilon^{ij}a^i b^j = a^1 b^2 - a^2 b^1, \qquad \epsilon^{12} = +1, \tag{10.1}$$

and $\mathsf{R}^{ij}(\vartheta) \equiv \cos\vartheta\,\delta^{ij} + \sin\vartheta\,\epsilon^{ij}$.

K^{ij} is a rotation. With the reflection $\Pi^{ij}(a) \equiv \delta^{ij} - 2a^i a^j/A$,

$$\Pi^{ik}(a)\,\Pi^{kj}(b) = \delta^{ij} - \frac{2a^ia^j}{A} - \frac{2b^ib^j}{B} + \frac{4\,(a\cdot b)\,a^ib^j}{AB} = 2\,C\,K^{ij}, \tag{10.2}$$

and the product of two reflections is the rotation by twice the angle between them:

$$K^{ij} = \frac{\Pi^{ik}(x-w)\,\Pi^{kj}(x-z)}{2\,(z-w)^2} = \frac{\mathsf{R}^{ij}(2\theta)}{2\,(z-w)^2}, \qquad u^iv^jK^{ij} = \frac{\cos 2\theta\; u\cdot v + \sin 2\theta\; u\times v}{2\,(z-w)^2}. \tag{10.3}$$

S^{ij}. From $S = K + \delta/2C + a^ib^j/(AB)$ and $\frac{1}{2}(1+\cos 2\theta) = \cos^2\theta$, $\frac{1}{2}\sin 2\theta = \sin\theta\cos\theta$:

$$S^{ij} = \frac{\cos\theta}{(z-w)^2}\,\mathsf{R}^{ij}(\theta) + \frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2}. \tag{10.4}$$

$\mathcal{A}_0T^{ij}$: the factor $1/\Delta$ cancels. T^{ij} vanishes when a and b are collinear; explicitly, with $\tilde a^i \equiv \epsilon^{ij}a^j$,

$$T^{ij} = \frac{a\times b}{C\,AB}\Big[(a\cdot c)\,a^i\,\tilde b^j - (b\cdot c)\,\tilde a^i\,b^j\Big], \tag{10.5}$$

and since $\Delta = |a\times b|$, with the signed angle $\vartheta_w \equiv \mathrm{sgn}(a\times b)\,\theta_w$:

$$\mathcal{A}_0\,T^{ij} = \frac{\vartheta_w}{(z-w)^2\,(x-w)^2\,(x-z)^2}\Big[(x-w)\cdot(z-w)\;(x-w)^i\,\widetilde{(x-z)}^{\,j} - (x-z)\cdot(z-w)\;\widetilde{(x-w)}^{\,i}\,(x-z)^j\Big]. \tag{10.6}$$

Row III in two dimensions.

$$\boxed{\mathcal{R}^{ij} = \frac{\mathsf{R}^{ij}(2\theta)}{2\,c^2}\left\{\log\frac{B}{A} + i\sigma\pi\right\} - \frac{1}{2}\left[\frac{\cos\theta}{c^2}\,\mathsf{R}^{ij}(\theta) + \frac{a^ib^j}{AB}\right]\log\frac{c^2}{A} + \frac{\vartheta_w}{c^2AB}\Big[(a\cdot c)\,a^i\tilde b^j - (b\cdot c)\,\tilde a^i b^j\Big]} \tag{10.7}$$

11 Step 9: one-parameter integral for the last two terms

Since $(a-tb)^2$ equals A at $t=0$ and C at $t=1$,

$$\log\frac{C}{A} = \int_0^1 dt\,\frac{d}{dt}\log(a-tb)^2 = \int_0^1 dt\,\frac{2\,(tB - a\cdot b)}{(a-tb)^2}, \tag{11.1}$$

and with (8.4) the last two terms of (1.2) combine to

$$-\frac{1}{2}\,S^{ij}\log\frac{C}{A} + \mathcal{A}_0T^{ij} = \int_0^1 dt\,\frac{T^{ij} - (tB - a\cdot b)\,S^{ij}}{(a-tb)^2} = \int_0^1 dt\,\frac{\mathcal{N}^{ij}(t)}{(z-w)^2\,\big(x_t-w\big)^2}, \tag{11.2}$$

$$\mathcal{N}^{ij}(t) = -(tB - a\cdot b)\,\delta^{ij} + \frac{A - 2a\cdot b + tB}{A}\,a^ia^j - (1-t)\,b^ib^j - \frac{t(A+B) - 2a\cdot b}{A}\,a^ib^j, \tag{11.3}$$

where $x_t = (1-t)x + tz$. (The coefficients follow from $c\cdot a + tB - a\cdot b = A - 2a\cdot b + tB$, $c\cdot b + tB - a\cdot b = -(1-t)B$, and $(a\cdot b)(A-B) + (tB - a\cdot b)(A+B) = B\,[t(A+B) - 2a\cdot b]$.)

12 Properties of the result

1. **No singularity at $(x-w)^2 = (x-z)^2$.** All poles of the original terms have cancelled and $\log(B/A)$ is regular there.

2. **Collinear configurations.** $\mathcal{A}_0 = \theta_w/\Delta$ diverges when w lies on the segment between x and z, but $T^{ij} \propto a\times b$ vanishes there and $\mathcal{A}_0T^{ij}$ stays bounded (it changes sign across the segment through ϑ_w).

3. **The imaginary constant.** The only remnant of the negative arguments is $i\sigma\pi K^{ij}$. With the prefactor i of (2.6) it is the real kernel $-\sigma\pi K^{ij}$:

$$\left.\frac{d^3N_{\mathrm{I.III}}}{d^3k}\right|_{\pi} = -\,\sigma\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\;u^i\,v^j\,K^{ij}\;\mathcal{W} + \text{c.c.} \tag{12.1}$$

4. **Reality.** Row III is X + c.c. and therefore real. For commuting ρ, $\mathcal{W}$ is real and

$$\frac{d^3N_{\mathrm{I.III}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^5}\,\frac{1}{k^+}\int u^iv^j\,\mathcal{W}\Big\{\sin\big[k\cdot(w'-w)\big]\Big(K^{ij}\log\frac{B}{A} - \tfrac{1}{2}S^{ij}\log\frac{C}{A} + \mathcal{A}_0T^{ij}\Big) - \sigma\pi\,\cos\big[k\cdot(w'-w)\big]\,K^{ij}\Big\}. \tag{12.2}$$

5. **Behaviour at $w \to z$ ($c \to 0$).** $K^{ij} \to \delta^{ij}/2C$, $S^{ij} \to \delta^{ij}/C$, while $K^{ij}\log(B/A)$ and $\mathcal{A}_0 T^{ij}$ are $\mathcal{O}(1/|c|)$:

$$\mathcal{R}^{ij} \;\longrightarrow\; -\frac{\delta^{ij}}{2\,(z-w)^2}\left\{\log\frac{(z-w)^2}{(x-w)^2} - i\sigma\pi\right\}. \tag{12.3}$$

   $\mathcal{W}$ does not depend on w and $\int d^2w\,\log(c^2)/c^2$ does not converge at $w \to z$. The source is t_{22} (the last term of N_{3b}) and the $i\sigma\pi$ term. So Row III as given is not finite by itself at $w \to z$.

6. **Relation to the other rows.** The piece $K^{ij}\log(B/A)$ has the same kernel as Group I Rows II and IV, Group II Row I (first part) and Group III Rows I, II; together these add up to commutators of ρ only (see the general note). What is specific to Row III is $i\sigma\pi K^{ij} - \frac{1}{2}S^{ij}\log(C/A) + \mathcal{A}_0 T^{ij}$.

13 Numerical checks

statement	check
$\sum_{n=1}^{22} t_n = KL_c - \frac{1}{2}SL_d + \mathcal{A}_0 T$	1500 random (a,b), complex principal-branch logs, relative deviation $< 10^{-10}$
coefficients of L_a and of L_b both equal $\mathcal{Q}^{ij}$	same run, 10^{-11}
$\mathcal{A}_0 = \theta_w/\Delta = \int_0^1 dt/(a-tb)^2$	same run; quadrature, 10^{-16}
two-dimensional forms of K, S, $\mathcal{A}_0 T$	same run, 10^{-9} or better
integral representation (11.2)	quadrature, 10^{-15}